

4. A ball of mass 3 kg has 750 J of potential energy when held at the top of a building. The ball is then dropped from rest.

- (a) (i) Use an equation from page 2 to calculate the height, h , of the building. (acceleration due to gravity, $g = 10 \text{ m/s}^2$) [3]

height, $h = \dots\dots\dots \text{ m}$

- (ii) When the ball hits the ground it has 600 J of kinetic energy. Use an equation from page 2 to calculate the velocity of the ball at this moment. [3]

velocity = $\dots\dots\dots \text{ m/s}$

- (iii) Use the information above to calculate the work done against air resistance as the ball falls. [1]

work done = $\dots\dots\dots \text{ J}$

- (iv) Use an equation from page 2 and your answers to parts (a)(i) and (iii) to calculate the mean force of air resistance during the fall. [3]

mean force = $\dots\dots\dots \text{ N}$



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(b) (i) Explain, in terms of named forces, why the ball reaches a terminal speed as it falls through the air. [3]

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(ii) Newton’s 3rd law of motion applies to the ball as it falls.
Complete the table to describe the Newton’s 3rd law force for each of the actions given. [2]

Action	Newton’s 3rd law force
Earth pulls downwards on the ball	
Ball pushes downwards against the air	

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